Water Hyacinth to agricultural manure: biomass estimation with Pléiades Neo optical imagery and aerobic composting in Puri, India

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***Abstract*—Water hyacinth infestation is a significant global environmental challenge affecting millions worldwide. Remedial actions include the removal of the plant or biocontrol, both of which have major disadvantages. As an alternative, we propose sustainable valorization of water hyacinth biomass through aerobic composting, supporting economic growth in the rural community while rejuvenating rural water bodies. However, scaling such an operation would require estimation of the collective biomass. Our proposition also includes the use of very high-resolution satellite imagery along with deep learning-based analytical techniques for biomass estimation that feed back to the cost analysis of the manure production at scale. Proof of the concept results show significant promise towards inclusivity and sustainable growth.**

***Index Terms*—Water Hyacinth, aerobic composting, biomass, segmentation, Pléiades Neo.**

I. Introduction

Water Hyacinth, *Pontederia crassipes* (formerly *Eichhornia crassipes*), is considered the world's most problematic aquatic weed due to its infestation of surface water bodies and particularly in the frost-free areas of the globe, which fall between 40°N and 30°S latitudes [1]. In the warm tropical and subtropical regions with abundant sunlight, water hyacinth mats propagate rapidly on the surface of the eutrophicated surface water bodies [2]. In India, the weed was introduced as an ornamental plant during the 1890s and has since spread across the country, infesting millions of ponds, lakes, canals and rivers. The infestation is particularly acute in states like Kerala, Odisha, and Assam, as well as in the dense human settlements [3] along the Indo-Gangetic plane.

Water hyacinth infestations are common in Southeast Asia and tropical countries like Sri Lanka. Major lakes and rivers in Africa, such as Lake Tana (Ethiopia), Lake Victoria, Lake Tanganyika, the Niger River etc. are suffering from water hyacinth infestation, highlighting the global scale of the challenge. The floating weed can reproduce both sexually as well as asexually, and its seeds remain viable in the benthic sludge for decades, making complete eradication very difficult and costly. This weed from the Amazonia helps it to outcompete native flora in infested water bodies for sunlight and nutrients, while decay of water hyacinth mats at the end of their lifecycle depletes the dissolved oxygen concentration, adversely affecting the native fauna [4]. The dense floating mat on the infested canals, lakes and rural ponds diminishes the potential of inland pisciculture as well as navigation [5]. Therefore, multiple large-scale attempts have been made through regular manual cleaning operations, biological or chemical control, or, lately, re-purposing the biomass into biodegradable arts and crafts materials or manure. The scope of chemical/bio control for this weed using herbicides is limited due to environmental concerns, while complete eradication is difficult to achieve using contemporary physical, chemical, or biological abatement techniques.

As an alternative, regular harvesting and utilisation of water hyacinth biomass offers an economically sustainable and environmentally benign solution. However, the large-scale implementation of such waste-to-wealth strategies suffers from a lack of scientific data critical for planning and decision-making. In addressing such a gap, this paper proposed the valorization of water hyacinth biomass through aerobic composting and its economic sustainability through pilot studies in the Puri, a coastal district of the Indian state of Odisha, India, where water hyacinth infestation has affected more than 2000 rural ponds, lakes and canals.

However, sustainable socioeconomic adaptation across large areas (district, state, or national level) would require estimating biomass through remote measurements, such as satellite imagery, drone surveys, in-situ IoT sensors, citizen science,

Acknowledgement: The work is supported by funding through a PhD studentship to Ms Aldhawi by the Government of Saudi Arabia; a project to ICRISAT sponsored by the Government of Odisha, India, under the State Plan: Innovation Scheme, Department of Agriculture and Farmers Empowerment; and a Newcastle University, UK, Global Partnership Fund.

or manual monitoring [1]. Out of these, only remote sensing through satellite imagery provides opportunities at scale. This enables monitoring the temporal and spatial distribution [6], as well as estimating the biomass of floating water hyacinth, which is critical to understanding the pattern of infestation, as the growth of water hyacinth is influenced by the geographic location of the infested water body as well as the nutrient level in the water [7]. Therefore, this work also proposes deep learning-based methods for estimating water hyacinth biomass using satellite imagery. For these purposes, we have utilised very-high-resolution Pléiades Neo optical imagery from Airbus at 30 cm resolution, which, to the best of the authors' knowledge, is the first of its kind for this application.

II. Background and Related Work

A. Remote sensing for water hyacinth monitoring

Satellites play a decisive role in detecting and monitoring water hyacinth [8–10]. Datta *et al.* [1] provided a review of possible methods to monitor water hyacinths, proposing the integration of multiple techniques such as satellites, radar, and machine learning; however, it did not include a real-world experiment. In [11], the first practical application of monitoring was presented using synthetic aperture radar (SAR) data from the Sentinel-1 satellite at Vembanad Lake, India. Authors in [12] utilised Sentinel-2 data to monitor the spread of water hyacinth from 2018 to 2021, employing the NDVI index as the primary classification tool. Authors in [13] use Sentinel-1 SAR data with statistical change detection algorithms, demonstrating the ability of radar to detect water hyacinths without relying on deep learning techniques.

On the contrary, techniques [14] were proposed using multispectral imaging using a drone and implemented a custom U-Net model for the segmentation of water hyacinth plants, achieving strong performance. However, the study was limited to low-altitude aerial images (10 meters), and the model was not tested on satellite images, which have greater spatial and spectral complexity, including lower image resolution and varying cloud cover. Similarly, [8] proposed a three-stage hierarchical classification approach (detection of water, aquatic plant, and water hyacinth) for monitoring based on free satellite data from Landsat-8 or Sentinel-2 via Google Earth Engine. However, the estimation of water hyacinth biomass from satellite imagery has yet to be fully explored.

B. Water hyacinth biomass and utilisation

Utilisation of water hyacinth biomass for making paper, handicrafts, packaging material, insulation boards, non-polar industrial mops, *etc.*, has been reported by several researchers [2]. The nutrient-rich water hyacinth, with high cellulose and hemicellulose content, is suitable for aerobic composting, particularly in rural contexts where there is no heavy-metal contamination [1]. Large-scale aerobic composting can be undertaken on a piece of land not prone to waterlogging and adjacent to the infested water body. The land requirement is dependent on the quantity of harvested water hyacinth biomass. Moreover, as water hyacinth biomass tends to rapidly lose its ammoniacal-nitrogen content through the release of ammonia in the ambient, within a 72 hour period after harvesting, the efficient utilisation of the biomass requires advanced planning regarding land requirements, hiring of labourers and other peripheral logistics. Accurate quantification of floating water hyacinth biomass, using satellite-based technologies, is hence intrinsically linked with optimisation of the costs associated with large-scale utilisation of the biomass for making aerobic compost. The same can be said about any value-added attempts using bulk quantities of this biomass.

Fig. 1. Example 0.3m resolution satellite images from Airbus Pléiades Neo used to train and test for the segmentation models.

C. Rural economics

The high moisture content ($>85\%$) of the water hyacinth biomass makes its transport expensive. As a result, any value addition proposition needs to be a techno-economically feasible decentralised model. The bulk quantities of the biomass harvested from the water bodies necessitate interventions at the scale of several hundred tons. While products such as garments or handicrafts can be made from water hyacinth, the process requires sophisticated skill development towards craftsmanship, market linkage and market demand. Utilisation of water hyacinth biomass for products such as compost, biochar or biogas, , which can be useful for the villagers, greatly enhances its scalability as well as replicability. Adequate awareness generation and capacity building of resource-poor rural women can encourage them to undertake value addition of water hyacinth biomass as a rural enterprise, without the need for elaborate forward linkage.

III. Study Area

Our larger study area encompasses multiple locations in Puri, a coastal district in the Indian state of Odisha, characterised by a tropical wet and dry or savanna climate (classification: Aw). The district is at sea level with an annual mean temperature is $29.04\,°\mathrm{C}$. Ground truthing of the work presented here was carried out in several villages of the Puri district. Biomass quantification and water quality assessment work was done for a community pond (latitude: 20.07778°N; longitude: 86.03115°E) located in the Bhilisasan village of the Nimapada block of Puri district, Odisha, India. For satellite image segmentation purposes, training data and related ground truth labels also include the extended area in Odisha.

IV. Methodology

Our overall approach consists of two main components: (A) estimating water hyacinth biomass using remote sensing techniques, and (B) converting the biomass into aerobic compost.

Fig. 2. Study area ground truth before and after harvesting using drone captures: column 1: 6^{th} December 2024 and column 2: 9^{th} April 2025.

A. Remote sensing-based biomass estimation

This paper proposes the use of satellite optical imagery in estimating water hyacinth biomass due to its coverage and historical archive. Our approach includes deep learning-based segmentation models, which lead to area calculation and, consequently, biomass estimation. However, this requires new training data, as no suitable data set exists for these purposes, especially in our study area. Therefore, our approach in this section is twofold: 1) creation of a data set and 2) biomass estimation model.

A custom dataset containing water bodies fully or partly covered with water hyacinth was created using very-high-resolution satellite images acquired from the Pléiades Neo satellite, provided by Airbus [15], at a spatial resolution of approximately 0.30m for an extended study area within the region. Example images from the dataset is shown in Fig. 1. The original images were divided into tiles of 512×512 pixels, and only the sections containing water hyacinths were selected, resulting in 347 annotated image tiles. The masks were created manually using QGIS software.

Consequently, the dataset is used to retrain an off-the-shelf, pre-trained segmentation algorithm suitable for satellite image segmentation. We chose the DeepLabv3 [16] model, with ResNet-50 [17] as its backbone. The DeepLabv3 is well-suited for high-resolution semantic segmentation tasks and has demonstrated strong performance in detecting complex object boundaries in satellite imagery. It contains Atrous Spatial Pyramid Pooling (ASPP) to capture richer contextual information which is essential for segmenting hard-to-identify aquatic plants, such as water hyacinth. In addition, an experiment was carried out using the U-Net [18] model for comparison. It is a popular model in the segmentation field and relies on an Encoder-Decoder architecture that enables it to recover spatial details through upsampling. Both models were trained and tested on the same dataset, using identical settings. The output of segmentation is a binary mask revealing the presence of water hyacinth. Based on the satellite image resolution, each segmented pixel corresponds to $0.09m^2$ area on the ground, and we correlated that with the biomass extracted from the water body, which was manually weighed for ground truthing. Empirically, this equates to 0.045 ton/m^2 or 45 kg/m^2.

B. Biomass conversion

Water hyacinth biomass was harvested using traditional methods by the villagers, which involved the use of bamboo and a fishing net. Total water hyacinth biomass harvested (refer Fig. 2) from the 1.37 ha pond was 287 tons, and the biomass was mixed with 82 tons of paddy straw and 41 tons of cow dung to prepare rectangular heaps of aerobic compost of about 1.5 m width and height each. Shredded paddy straw and water hyacinth biomass were mixed with cow dung to prepare multiple rectangular-shaped compost heaps as per the optimal biomass ratio mentioned in the section above. For every 1 ton of mixed biomass, 1 kg of microbial culture was used by mixing it with cow dung as a slurry. Biomass quantities were weighed using an electronic platform weighing balance with digital display (Make: Smart, Model: LAE66 PLAX, T-10-100; Capacity: 100 kg; accuracy: 10 gm).

Fig. 3. Harvesting and utilisation of water hyacinth biomass for making aerobic compost.

Appropriate quantities of each biomass were mixed to make horizontal stratified layers one upon the other, creating several heaps as shown in Fig. 3. The length of the heap was as per the availability of the land area. Water samples were collected from the pond in the study area and analysed for physical and chemical characteristics [19]. Manual heap turnover and watering were carried out at an interval of 12 days. Heap temperatures were monitored using a digital thermometer. The moisture content of different biomass and compost was measured with a digital moisture meter (Vunexo, VU-SM-809). The final compost was prepared and sieved using a 4mm electrically operated sieving machine (Make: ALTECH Construction Equipment Ltd., 1 HP electrical motor, length: 2.4 m, width: 0.6 m) and packed in synthetic gunny bags.

V. Results and Discussion

The proposed DeepLabv3 model for segmentation was trained on our satellite imagery dataset for 35 epochs using the Adam optimizer algorithm with a learning rate of $1e-4$. The data was divided into 70% training, 15% validation, and 15% testing sets. All training and evaluation experiments were conducted using Google Colab Pro with an NVIDIA T4 GPU, utilizing Python and the PyTorch framework to accelerate training and enhance model execution efficiency.

The results achieved 0.5997 mean Intersection over Union (IoU) across all test samples. This result is a good indication

Fig. 4. Example of segmentation results showing the satellite image (left) captured by Pléiades Neo on 5^th April 2025 for our study area, ground truth mask (middle), and predicted mask using the segmentation model (right).

of the model's ability to identify water hyacinth areas with acceptable accuracy, with some samples achieving high accuracies exceeding 0.9, reflecting the model's robustness in clear-cut cases. For comparison, we also used U-Net segmentation model on the same dataset and with the same training settings. However, the model demonstrated poorer performance, with an average IoU on the test set of 0.3985, reflecting its limited ability to distinguish diffusion regions accurately. Therefore, it was not adopted in the final results.

These results demonstrate its ability to identify water hyacinth distribution areas with reasonable accuracy under high-density and spectral clarity conditions; however, it exhibits poorer performance in areas with sparse distribution, cloud shadows, or high water turbidity. This suggests the need to enhance the training database by including diverse labelled samples and various imaging conditions. Alternative models, such as SegNet [20] or Transformers [21], could also be tested in the future to enhance the ability to capture fine plant boundaries in complex aquatic environments.

Figure 4 shows a visual example of the model's segmentation outcome for an image captured on 5^{th} April 2025, showing the original satellite image on the left, the ground truth mask in the middle, and the predicted mask on the right. The example demonstrates that the model was able to identify most of the classified regions as in the ground truth data, with some additional regions detected or small portions missed, indicating good overall classification performance with room to improve prediction accuracy and reduce these minor errors.

In this segmented sample, the number of pixels classified as water hyacinth was also calculated, reaching $18,026$ pixels. This number was then converted to area using the pixel size, resulting in a total estimated area of 1622.34 m^2. These calculations help accurately estimate the plant's distribution and actual area on the ground, and hence the biomass.

For biomass conversion, samples were collected from each heap of compost prepared for this bulk production, with an average C/N ratio of 13.87 and organic carbon content of 13.21 mg/kg. Samples of the compost prepared were collected and sent for physico-chemical analysis. The analytical data presented in Table I highlights that the compost prepared was in compliance with the FCO standard.

TABLE I
QUALITY OF COMPOST PREPARED (WITH FERTILISER CONTROL ORDER GUIDELINES)

Parameter	Unit	Compost Prepared	FCO standard
pH		7.23	6.5–7.5
Colour		Black	Dark brown or black
Odour		Absent	Absent
Moisture content	% by weight	18.2	15–25 (City compost) 15–20 (Vermicompost)
C/N ratio		13.87	¡ 20
OC %	mg/kg	13.21	12 (City compost) 18 (Vermicompost)
Total-N	mg/kg	7934	8000 (City compost) 10000 (Vermicompost)
Total-P	mg/kg	3874	4000 (City compost) 8000 (Vermicompost)
Total-K	mg/kg	5023	4000 (City compost) 8000 (Vermicompost)
Total-Ca	mg/kg	10343	
Total-Mg	mg/kg	3875	
Total-Fe	mg/kg	3244	
Total-Cu	mg/kg	2	
Total-Mn	mg/kg	234	
Total-S	mg/kg	1874	
Total-Zn	mg/kg	56	1000
Total-B	mg/kg	12	
Total-Na	mg/kg	1124	0

Socio-economic impact: Approximately 140 tons of aerobic compost were prepared using the water hyacinth biomass harvested from a 1.37 ha pond over a total period of less than 70 days. Each step of the process, viz. harvesting, heap preparation, heap maintenance, sieving and transport of compost to agriculture fields, generated local rural employment opportunities. The cost of production, calculated considering the cost of raw materials and labour, was found to be about ₹8 per kg and a conservative sale price of ₹10 per kg indicated a net profit potential of ₹$280,000$ or about $3000. Such an alternative localised livelihood can transform the lives of resource-poor women who took active part in this work. Cleaning of water hyacinth, in turn, increases inland pisciculture-related livelihood opportunities.

VI. CONCLUSION

Water hyacinth is considered one of the most challenging aquatic weeds, commonly found in tropical regions and experiencing major infestations across India. Current abatement techniques are inadequate. This paper proposes the creation of aerobic compost from the biomass to foster alternative livelihood opportunities in rural India. A biomass estimation technique complements this proposition through very high-resolution (30cm) satellite image acquisition and a state-of-the-art segmentation model. The results and analysis show the viability of the idea and promise for future growth.

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